

EXPERIMENT-6

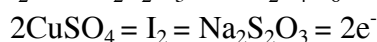
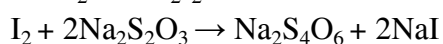
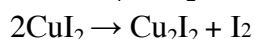
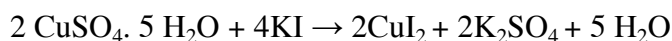
AIM: To determine the amount of Cu in the copper ore solution provided hypo solution(0.005 N).

Apparatus Required: Funnel, Burette, titration flask, Pipette, Stand.

Chemicals Required: Hypo solution (.005 N), KI, dil. Acetic acid, Na₂CO₃, H₂SO₄, CuSO₄, Starch & distilled water.

Theory: This experiment comes under the category of Iodometry. A known excess of solid KI is added to

Chemical reactions:



[Equivalent wt. of Cu²⁺ = 2 x 63.5/2 = 63.5]

Procedure:

1. Rinse & fill the burette with 0.005 N Na₂S₂O₃ solution.
2. Measure 10 mL of CuSO₄ solution in a clean conical flask.
3. Neutralise any free acid present by adding Na₂CO₃ solution drop by drop till a faint permanent precipitate remains even on shaking. [Cu₂(OH)₂ CO₃], basic copper carbonate is formed].
4. Add dil. Acetic acid dropwise until the precipitate just dissolves.
5. Add 5 mg of solid KI, over the mouth of the conical flask by filter paper and allow the mixture to stand for 2-5 minutes in the dark. The solution becomes brown due to liberated iodine.
6. Titrate the liberated iodine with hypo solution added from the burette; the brown colour of iodine becomes fainter as the addition of hypo solution proceeds, and when only a faint yellow colour remains add 1 drop of starch solution. This immediately forms a deep blue iodio-starch complex.
7. Add hypo solution further drop by drop till the blue colour of the solution just disappears.
8. Repeat the titration to get a set of three readings.

Observation:

S.No.	Volume of water sample taken (ml)	Burette reading		Concordant Volume of Hypo used(ml)
		Initial	Final	
	10			
1		0	5.5	5.5
2		5.5	11.0	5.5
3		10.5	15.7	5.2

Calculation:

(CuSO₄) (Na₂S₂O₃)

$$N_1 V_1 = N_2 V_2$$

$$N_1 = N_2 V_2 / V_1$$

$$= 0.005 \times V_2 / 10$$

$$= 0.0005 V_2$$

Strength of CuSO₄ solution = N₁ X Equivalent weight(63.5gms) =
.....mg/L.

Result: The Strength of CuSO₄ solution =